Name: ______________________ **Worksheet**

Using the cosine ratio

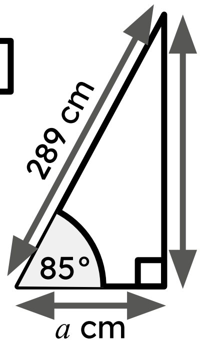
Task A: Ratio tables, the unit circle, and tangents

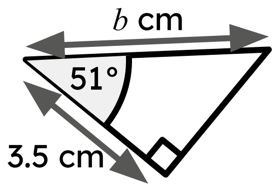
1) Substitute the labelled sides and angle in each triangle into the given version of the cosine formula, and find the length of its missing side.

$$h \times \cos(\theta°) = \text{adj}$$

$\boxed{289} \times \cos(\boxed{}) = \boxed{}$

$a = \boxed{}$





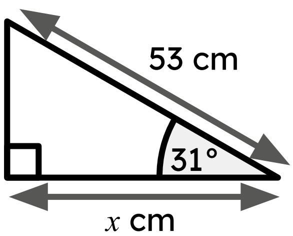
$$\frac{\text{adj}}{\cos(\theta°)} = h$$

$\dfrac{\boxed{}}{\cos(\boxed{})} = \boxed{}$

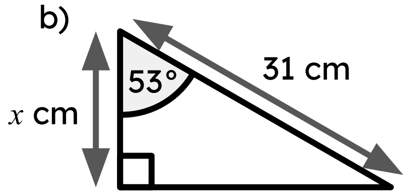
$b = \boxed{}$

2) Match the triangle to the equation showing the relationship between its hypotenuse and an adjacent side.
Use the equation to find the length of the missing side labelled x on each triangle (2 d.p.)

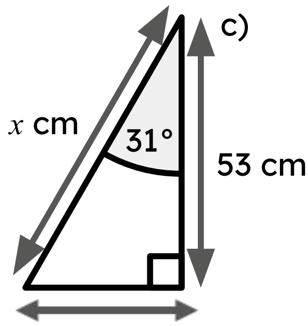
a)



b)



c)

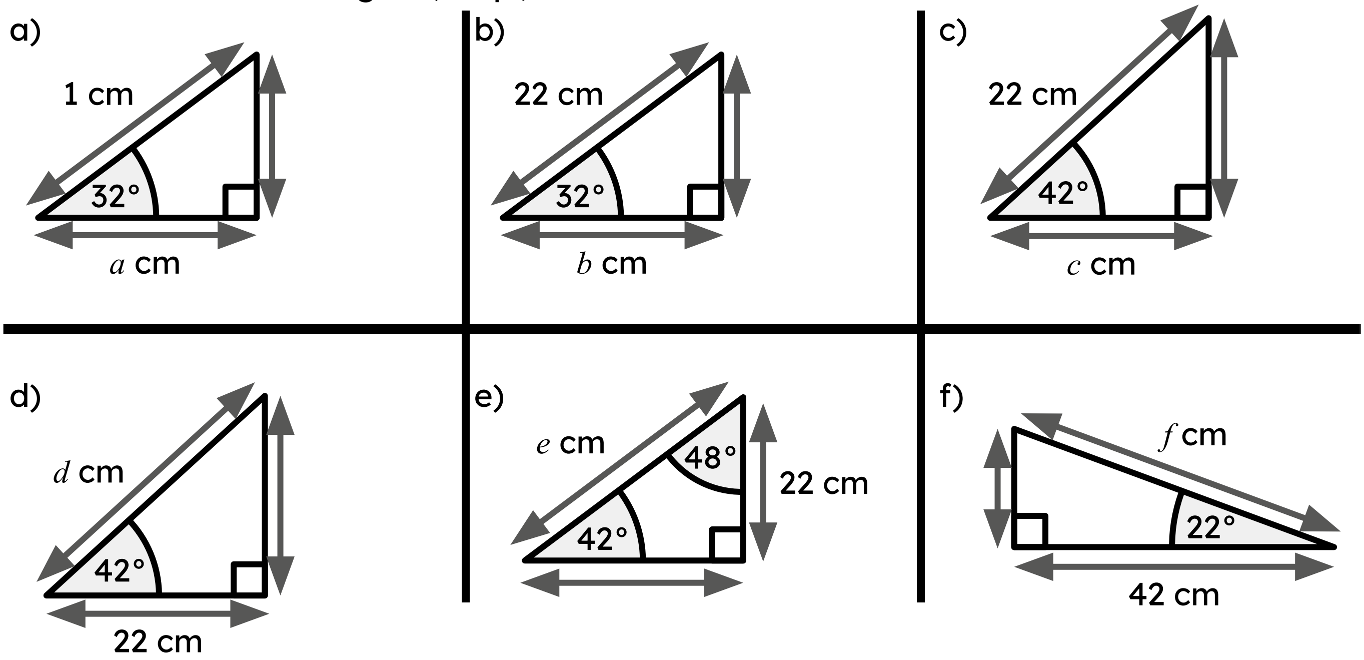


d $53 \times \cos(31°) = x$

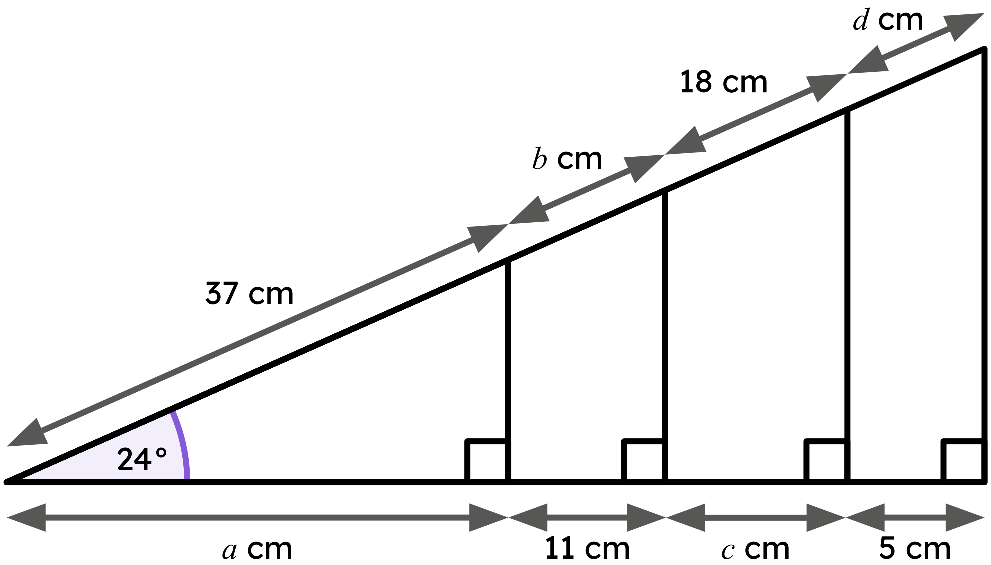
e $x \times \cos(31°) = 53$

f $31 \times \cos(53°) = x$

3) By using the cosine formula, find the length of the missing side labelled with a letter for each of these triangles (2 d.p.)

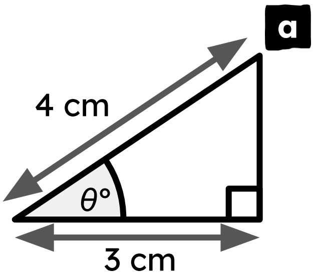


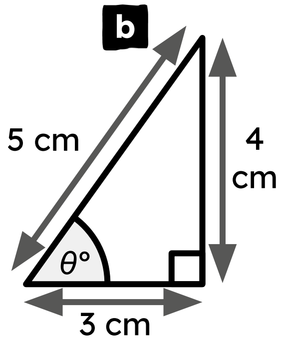
4) Find the lengths of the distances a, b, c, and d (rounded to the nearest integer).

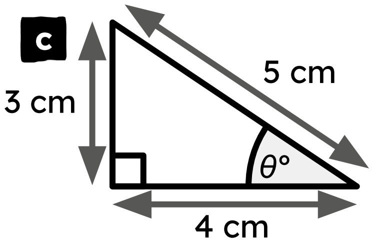


Task B: Calculating angles using the cosine formula

1) Match each triangle to: an equation that describes the relationship between its hypotenuse and an adjacent side, and to the size of its angle (2 d.p.).







d $\cos(\theta°) = \frac{3}{4}$

e $\cos(\theta°) = \frac{4}{5}$

f $\theta° = \cos^{-1}\left(\frac{3}{5}\right)$

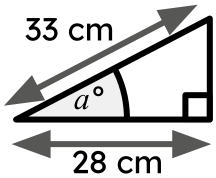
g $\theta° = 53.13°$

h $\theta° = 36.87°$

i $\theta° = 41.41°$

2) Find the size of each angle marked with a letter (1 d.p.)

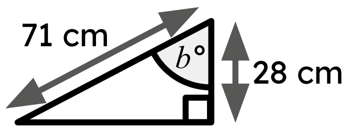
a)



$\cos(a°) = \frac{28}{33}$

$a° = \cos^{-1}\left(\frac{28}{33}\right) = \boxed{}$

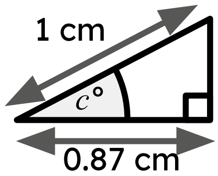
b)



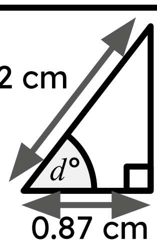
$\cos(b°) = —$

$b° = \cos^{-1}(—) = \boxed{}$

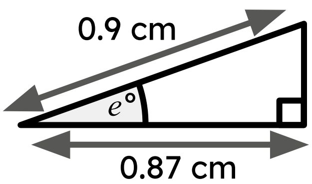
c)



d)



e)



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